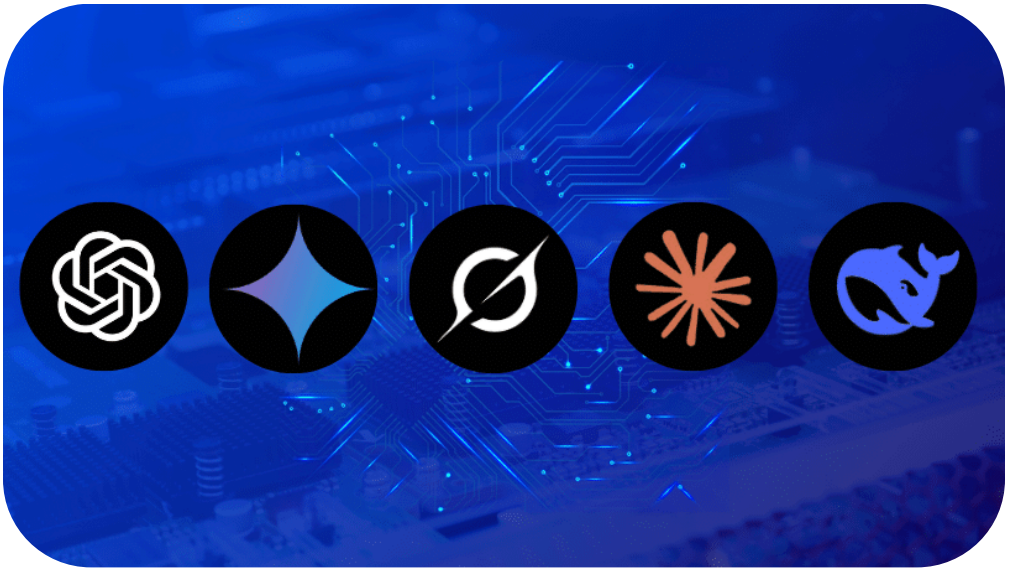
🤩 Ultimate model guide.



Quick Decision Matrix

Priority	Best Models	Why Use	Cost Range
Top Performance	GPT-5, Claude Opus 4	Best reasoning, coding, long-context	High ($15-75/1M tokens)
Balanced	Mistral Medium 3, Claude Sonnet	Good quality, reasonable cost	Medium ($0.40-2/1M tokens)
Speed + Cheap	Gemini Flash, DeepSeek	High throughput, low cost	Low ($0.04-0.50/1M tokens)
Multi-Model Access	OpenRouter	One API, many models, fallbacks	Variable (depends on model)

Model Details & When to Use

Premium Models (High Performance)

- **GPT-5** - 256K context, excellent reasoning, 3 variants (standard/mini/nano)
- **Claude Opus 4** - Superior coding, can work autonomously for 7+ hours

- **Use for**: Complex AI agents, advanced reasoning, critical applications

Balanced Options (Sweet Spot)

- **Mistral Medium 3** - Near-premium performance at lower cost
- **Claude Sonnet 3.5** - Good all-around performance
- **Use for**: General applications, balanced cost/performance needs

Fast & Cheap (High Volume)

- **Gemini Flash** - ~314 tokens/sec, very responsive
- **DeepSeek** - Ultra-low training costs, open weights
- **Llama 3.2 1B** - ~558 tokens/sec, extremely budget-friendly
- **Use for**: High-volume tasks, drafts, simple queries

When to Use OpenRouter

OpenRouter is perfect when you need:

✅ **Model Shopping** - Test 400+ models with one API
✅ **Automatic Fallbacks** - If one provider fails, auto-switch to backup
✅ **Cost Optimization** - Easily compare prices across providers
✅ **Unified Interface** - Same schema for all models (OpenAI-compatible)
✅ **Provider Flexibility** - Access Google, Anthropic, Mistral, DeepSeek, etc.

Best for: Production apps that need reliability, cost control, and easy model switching.

Key Considerations for AI Agents

- **Context Window**: GPT-5 (256K), Claude (200K+) for long conversations
- **Tool Use**: OpenAI and Anthropic excel at function calling
- **Persistence**: Claude Opus 4 maintains coherence over extended tasks
- **Cost at Scale**: Consider open-weight models (DeepSeek, Llama) for high volume

Pro Tips

1. **Benchmark with real prompts** - Use tools like Helicone to monitor performance
2. **Start cheap, scale up** - Begin with Gemini Flash or DeepSeek for testing
3. **Use fallbacks** - Primary model → backup → cheap batch processing
4. **Monitor costs** - Track token usage, especially with premium models